Last Name: First Name:

Matriculation Number:

Exam
Artificial Intelligence 1

April 9, 2025

Please ignore the QR codes; do not write on them, they are for grading support

	To be used for grading, do not write here												
prob.	1.1	2.1	2.2	3.1	4.1	5.1	5.2	5.3	6.1	7.1	7.2	Sum	grade
total	10	8	10	7	8	6	8	6	8	12	7	90	
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In the course Artificial Intelligence I/II we award bonus points for the first student who reports a factual error in an old exam. (Please report spelling/formatting errors as well.)

1 Prolog

Problem 1.1 (Interpreting and Writing Clauses)
Consider the following sequence of Prolog clauses:

```
descendant(D,X) :- descendant(D,P), parent(P,X).
descendant(D,X) :- parent(D,X).
sibling(X,Y) :- parent(P,X), parent(P,Y).
sibling(william,harry).
parent(elizabeth,charles).
parent(charles,william).
```

Below, you may abbreviate all names by their first letter.

1. Give all function and predicate constants and their arities that make up the first-order signature underlying these clauses. 4 pt

   Solution: nullary function constants: elizabeth, charles, william, harry
   binary predicate constants: sibling, parent, descendant

2. Give a clause that can be used to extend the above with the definition of a binary predicate for the niece-or-nephew relation. 2 pt

   Solution: `nieceOrNephew(X,N) :- sibling(X,P), parent(P,N).`

3. Interpreting the clauses as a Prolog program, what is output when we execute the query `descendant(X,Y)`? 2 pt

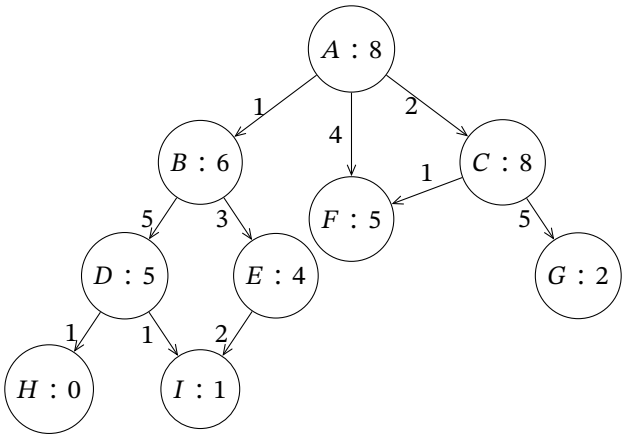
   Solution: Infinite recursion without any results

4. Interpreting each clause as a hypothesis in first-order predicate logic, which instances of the formula `descendant(X,Y)` can be derived? 2 pt

   Solution: `descendant(elizabeth,charles)`
   `descendant(elizabeth,william)`
   `descendant(charles,william)`

2 Search

Problem 2.1 (Search Algorithms)
Consider the graph below. Every edge is labeled with its cost.
Each node is labeled with $n : h(n)$ where n is its name and $h(n)$ its value for heuristic h.



Assume you have already expanded the root node A. In each problem below, list the **next** 4 nodes that will be expanded using the respective search algorithm and (if needed) heuristic h.
If there is a tie, break it using alphabetical order.

1. depth-first search 1 pt

   Solution: B, D, H, I

2. breadth-first search 1 pt

   Solution: B, C, F, D

3. uniform-cost search 2 pt

   Solution: B, C, F, E

4. greedy search 2 pt

   Solution: F, B, E, I

5. A^*-search 2 pt

   Solution: B, E, I, F

Problem 2.2 (Search Problems)
Fix the set $V = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.
Consider the family of deterministic search problems $P_n = \langle S, A, T, I, G_n \rangle$ for $n = 0, 1, 2, ...$ given by

- S is the set of finite sequences over V.
- $A = \{b\} \cup \{a_i \mid i \in V\}$

- T is given by
  - $T(a_i, s) = \{s'\}$ where s' is s with i appended at the end
  - $T(b, s) = \begin{cases} \emptyset & \text{if } s \text{ is the empty sequence} \\ \{s'\} & \text{otherwise, where } s' \text{ is } s \text{ without its last element} \end{cases}$
- I is the set containing the empty sequence.
- G_n is the set of all sequences of distinct elements that sum to n.
  For example, G_3 contains (among others) 3 and 0, 3 and 2, 0, 1 but not 1, 1, 1 (whose elements are not distinct) or 1, 3 (whose elements do not sum to 3).

1. Give the result of applying the action sequence a_0, a_1, b, a_2 in the initial state. 2 pt

   Solution: 0, 2

2. Assume BFS tries the actions in the order $a_0, \ldots, a_9, b$. Which solution will it find first for P_{22}? 3 pt

   Solution: a_5, a_8, a_9 (Note that BFS ends up trying paths in ascending order of the corresponding numbers if each a_i is read as the digit i.)

3. Give a value of n for which P_n has no solution. 2 pt

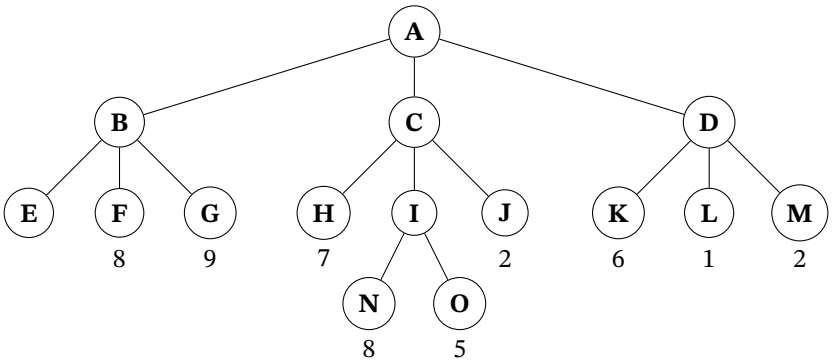
   Solution: Any $n > 45$

4. First explain the key reason why informed search strategies will typically perform better on this particular problem than uninformed ones; then give a useful heuristic for P_n. 3 pt

   Solution: Many states are detours: because they have a repetition or too large a sum, only going backwards via action b can lead to a goal. A heuristic can detect this right away and abandon that branch of the search tree.
   E.g., $h(s) = \infty$ if s has a duplicate element or if $\Sigma s > n$, and $h(s) = n - \Sigma s$ otherwise. (This heuristic is not admissible, but useful and admissible are not the same thing.)

3 Adversarial Search

Problem 3.1 (Minimax)
Consider the following minimax game tree for the **maximizing** player's turn. The values at the leaves are the static evaluation function values of those states; some of those values are currently missing.



1. Label the nodes I and C with their minimax value. 2 pt

Solution: I: 8, C: 2

2. Give every possible label for the node **E** that would result in the player definitely choosing move **B** (no matter how ties are broken). 2 pt

Solution: Any label > 2

3. Now assume **E** is labeled with 5, and we use $\alpha\beta$-pruning. We expand child nodes in alphabetical order. Which nodes would be pruned? 3 pt

Solution: O, M

4 Constraint Satisfaction

Problem 4.1 (Basic Definitions)
Consider the following constraint network $\langle V, D, C\rangle$

- $V = \{a, b, c, d\}$
- $D_a = \texttt{bool}$, $D_b = D_c = \{0, 1, 2, 3\}$, $D_d = \{0, 1, 2, 3, 4, 5, 6\}$
- Constraints C:
  - if a, then $b \leq 2$
  - if $c < 2$, then a
  - $b + c < 4$
  - $b > d$
  - $d = 2c$

1. Give all solutions. 1 pt

Solution: There are 2 solutions: a true, $b \in \{1, 2\}$, $c = 0$, $d = 0$

2. Give an inconsistent total variable assignment. 1 pt

Solution: Any total variable assignment that is not a solution, (see previous question) e.g., a false, $b = 0, c = 0, d = 0$

3. Give all consistent partial assignments α such that $\mathrm{dom}(\alpha) \subseteq \{a, b\}$. 0 pt

Solution: We classify the possibly assignments a by their domain:

- $\mathrm{dom}(\alpha) = \emptyset$: 1 assignment, namely α undefined everywhere
- $\mathrm{dom}(\alpha) = \{a\}$: 2 assignments, namely $\alpha(a) \in D_a$, undefined elsewhere
- $\mathrm{dom}(\alpha) = \{b\}$: 4 assignments, namely $\alpha(b) \in D_b$, undefined elsewhere
- $\mathrm{dom}(\alpha) = \{a, b\}$: 7 assignments, namely
  - 4 assignments with $\alpha(a) = false$, $\alpha(b) \in D_b$, undefined elsewhere
  - 3 assignments with $\alpha(a) = true$, $\alpha(b) \in \{0, 1, 2\}$, undefined elsewhere

4. Assume a CSP algorithm starts by assigning $true$ to a. How do the domains of b, c, d change after forward checking? 1 pt

Solution: Only D_b changes, namely to $\{0, 1, 2\}$

5 Logic

Problem 5.1 (Propositional Logic)

1. Fix the signature with propositional variables p, q, r. Give a formula that has exactly 1 satisfying assignment. 2 pt

Solution: E.g., $p \wedge q \wedge r$

2. Explain the Unit Propagation rule that is used in DPLL solvers for SAT. 2 pt

Solution: If we have derived p (resp. $\neg p$), replace every occurrence of p in other derivations with T (resp. F).

3. Assume C is in CNF, explain how you can get a formula in DNF that is equivalent to $\neg C$. 2 pt

Solution: Start with C and swap all $\wedge$ and $\vee$ with each other, and negate every literal.

Problem 5.2 (Predicate Logic)
Consider the following signature of predicate logic:

- unary function constant f
- binary function constant g
- binary predicate constant p

1. Give a formula that uses all three constants and is a theorem. 2 pt

Solution: E.g., $A \Rightarrow A$ where $A = \forall x.p(f(x), g(x,x))$.

2. Now consider the formulae A and B given by 3 pt

$$A = \forall x.\forall y.p(f(x), g(x,y))$$
$$B = \forall x.\forall y.p(x,y)$$

Give a model $\langle \mathcal{D}, \mathcal{I} \rangle$ that satisfies A but falsifies B.

Solution: E.g., $D = \{0,1\}$, $\mathcal{I}(f)(x) = 0$, $\mathcal{I}(g)(x,y) = 0$, $\mathcal{I}(p) = \{(0,0)\}$.

3. Prove or refute the following statement: 3 pt
If the formula A has one free variable x, and there is a model in which A is true under all assignments, then A is valid.

Solution: False. A counter-example is given by $G = p(x,x)$ and any model $\langle \mathcal{D}, \mathcal{I} \rangle$ in which $\mathcal{I}(p) = D \times D$.

Problem 5.3 (Proving)
Consider first-order predicate logic with unary predicate symbols P and Q.

1. Prove the following formula using tableaux calculus: 6 pt

$$(\forall x.P(x) \Rightarrow \neg Q(x)) \Rightarrow ((\exists y.P(y)) \Rightarrow \neg \forall z.Q(z))$$

Solution:

$$\frac{((\forall x.P(x) \Rightarrow \neg Q(x)) \Rightarrow ((\exists y.P(y)) \Rightarrow \neg(\forall.Q())))^{\mathsf{F}}}{(\forall x.P(x) \Rightarrow \neg Q(x))^{\mathsf{T}}(1)}$$
$$((\exists y.P(y)) \Rightarrow \neg(\forall z.Q(z)))^{\mathsf{F}}$$
$$(\exists y.P(y))^{\mathsf{T}}(2)$$
$$\neg(\forall z.Q(z))^{\mathsf{F}}$$
$$(\forall z.Q(z))^{\mathsf{T}}(3)$$
$$P(s)^{\mathsf{T}} \text{ (from (2) for fresh } s)$$
$$Q(s)^{\mathsf{T}} \text{ (from (3) with } s)$$
$$(P(s) \Rightarrow \neg Q(s))^{\mathsf{T}} \text{ (from (1) with } s)$$

$P(s)^{\mathsf{F}}$	$\neg Q(s)^{\mathsf{T}}$
	$Q(s)^{\mathsf{F}}$
close via $P(s)$	close via $Q(s)$

6 Knowledge Representation

Problem 6.1 (ALC)
Consider the following description logic signature

- concept symbols: m (for male), f (for female)
- role symbols: p (for parent-of), s (for sibling-of)

It is meant to model the world as indicated in the brackets.

1. Give an ABox for this signature that declares a woman E, her son C, and his sons W and H. 2 pt

   Solution: $E : f, C : m, W : m, H : m, EpC, CpH, CpW$

2. Give a concept describing the set of childless men who have a niece. 2 pt

   Solution: $m \sqcap (\forall p.\bot) \sqcap \exists s.\exists p.f$

3. ALC is not good at expressing numeric constraints such as forming the concept of people with at least **two** children. Assume we add a production $F ::= \exists^n R.F$ to the grammar that behaves like $\exists$ but requires at-least-n elements instead of at-least-one. 2 pt
   Give an appropriate definition of the interpretation of $\exists^n R.F$ in a model $\langle \mathcal{D}, [\![-]\!] \rangle$.

   Solution:
   $$[\![\exists^n R.F]\!] = \{x \in \mathcal{D} \mid n \leq \#(\{y \in \mathcal{D} \mid (x,y) \in [\![R]\!], y \in [\![F]\!]\})\}$$

4. Give the translation to first-order logic of the axiom $(\forall p.f) \sqsubseteq (\exists s.m)$ 2 pt

   Solution:
   $$\forall x.((\forall y.p(x,y) \Rightarrow f(y))) \Rightarrow (\exists y.s(x,y) \wedge m(y))$$

7 Planning

Problem 7.1 (Planning Deliveries in STRIPS)
Consider a truck that can carry 4 **objects at a time** and is supposed to deliver objects $Obj = \{V, W, X, Y, Z\}$ from location A to certain locations $Loc = \{A, B, C, D\}$ along some roads

$$Roads = \{(A,B),(B,A),(B,C),(C,B),(B,D),(D,B)\}.$$

We use the following STRIPS task:

- facts: $\{\texttt{at}(l,o) \mid l \in Loc, o \in Obj\} \cup \{\texttt{truck}(l) \mid l \in Loc\}$
- the action $\texttt{move}(l,m,O)$ delivers up to 4 objects from l to m: for $(l,m) \in Roads$ and $O \subseteq Obj$, and $\#(O) \leq 4$, it is given by
  - precondition: $\texttt{at}(l,o)$ for all $o \in O$, $\texttt{truck}(l)$

  - add list: $\texttt{at}(m, o)$ for all $o \in O$, $\texttt{truck}(m)$
  - delete list: same as precondition
- initial state: $\texttt{truck}(C)$, $\texttt{at}(A, o)$ for $o \in Obj$
- goal state: $\texttt{at}(C, V)$, $\texttt{at}(D, W)$, $\texttt{at}(B, X)$, $\texttt{at}(B, Y)$, $\texttt{at}(A, Z)$

1. Give a sequence of two actions that is applicable in the initial state and give the resulting state. 4 pt

   Solution: $\texttt{move}(C, B, \emptyset), \texttt{move}(B, Q, \emptyset)$ where $Q \in \{A, C, D\}$ results in $\texttt{truck}(Q)$, $\texttt{at}(A, o)$ for $o \in Obj$

2. Give an optimal plan for the task above. 4 pt

   Solution: $\texttt{move}(C, B, \emptyset), \texttt{move}(B, A, \emptyset), \texttt{move}(A, B, \{V, W, X, Y\}), \texttt{move}(B, C, \{V\}), \texttt{move}(C, B, \emptyset), \texttt{move}(B, D, \{W\})$

3. Consider the following heuristic: $h(s) = \frac{1}{4} \sum_{o \in Obj} d(s, o)$ where $d(s, o)$ is the number of roads separating the location of o in state s from its location in the goal state. Argue whether this heuristic is admissible or not. 4 pt

   Solution: The heuristic is admissible. An optimal plan must make at least $d(s, o)$ steps to move object o to its goal state, so $\sum_{o \in Obj} d(s, o)$ steps in total. At best, it can carry 4 objects in each step. So $h(s)$ is smaller than the minimal number of steps in any optimal plan (which is the definition of admissible).

Problem 7.2 (Search as Planning)
We saw in the course that planning is a special case of search. Complete the argument below establishing that the converse is also true.

1. Given a search problem $\langle S, A, T, I, G \rangle$, we define a planning task $\langle P', A', I', G' \rangle$ as follows: 7 pt

   - P' contains the facts $start$ and $inGoal$ as well as one fact $in(\boxed{s})$ for every $s \in \boxed{S}$.
   - A' contains the following actions, all given as triples $\langle pre, add, del \rangle$ of preconditions, add-list, and delete-list:
     - $init(s)$ of the form $\langle \{start\}, \{in(s)\}, \{start\} \rangle$ for every $s \in I$
     - $goal(s)$ of the form $\langle \{in(s)\}, \{inGoal\}, \{in(s)\} \rangle$ for every $s \in G$
     - $do(s, a, s')$ of the form $\langle in(s), \boxed{in(s')}, \boxed{in(s)} \rangle$ for every $s \in S$, every $a \in A$, and every $s' \in \boxed{T(s,a)}$
   - $I' = \{start\}$
   - $G' = \{inGoal\}$

   Now we have a bijection between

   - solutions $a_1, \ldots, a_n$ (visiting the states $s_0, \ldots, s_n$) for the search problem, and

- plans of the form

$$init(s_0), do(s_0, a_1, s_1), \ldots, do(s_{n-1}, a_n, s_n), goal(s_n)$$

that satisfy the planning task.

Solution: filled in above
